Extraction or Displacement? Labor Institutions and Surplus Distribution in India's Textile Value Chains

Master of Arts in Public Policy Dissertation Defense

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The Problem & Research Question

The Context

India's Textile & Apparel sector employs nearly 45 million people, but over **80%** work in the informal, unorganized sector. As formal lead firms modernize and increase productivity, what happens to these millions of informal workers?

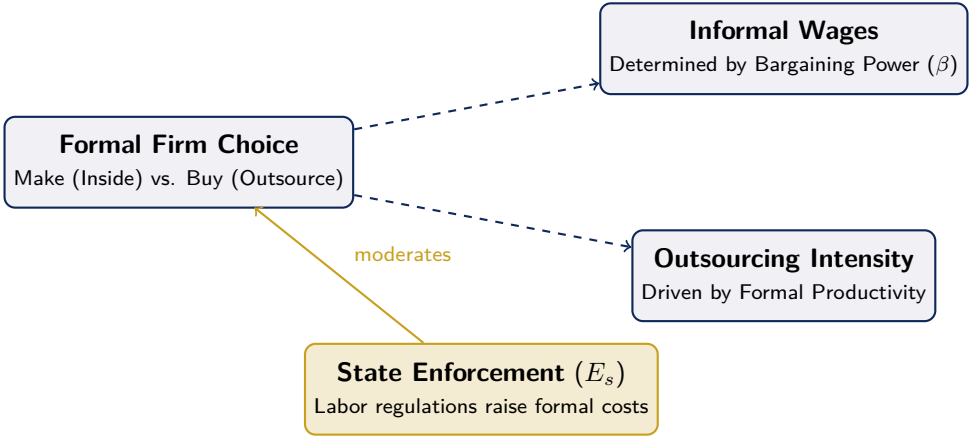
The Core Question

Is informality a *dying vestige* of underdevelopment, or a *structural feature* of modern capitalism?

Two Competing Hypotheses

- **Displacement:** Formal modernization naturally absorbs informality as the economy develops (Lewis, 1954).
- **Adverse Incorporation:** Formal firms actively exploit informality as a structural subsidy, permanently locking them in (Sanyal, 2007).

The Theoretical Framework



The Mechanism: A CES-Nash bargaining framework formalizes the "Make vs. Buy" decision. As formal productivity (A_f) rises, does the firm share rents via higher informal wages, or strategically offshore tasks to avoid regulated labor?

Data & Methodology

Integrating Surveys (2010–2024)

- **Formal Sector:** Annual Survey of Industries (ASI)
- **Informal Sector:** NSS/ASUSE unorganized enterprises
- **Matched Panel:** $N = 152$ state-industry-year observations (28 states, Textiles & Apparel)
- **Enforcement (E_s):** State-level factory inspection ratios

Methodological Approach

- **High-Dimensional Fixed-Effects:** State clustered standard errors with Year and NIC controls.
- **Focus on Reduced-Form Tests:** Rather than highly parameterized structural estimation, the analysis relies on robust, falsifiable comparative statics.
- **Rigorous Diagnostics:** Oster bounds for OVB, Breusch-Pagan, Shaprio-Wilk, and 10 robustness checks.

Finding 1: Strategic Outsourcing

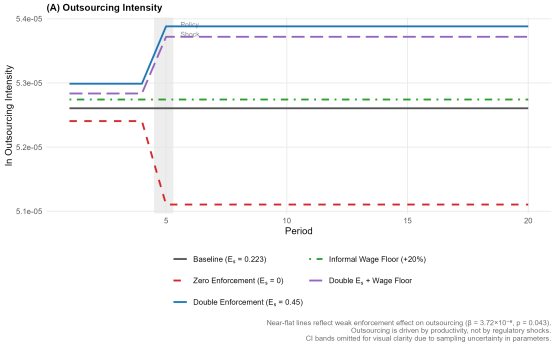
Regulatory Arbitrage

Result: Formal productivity correlates positively with informal outsourcing intensity ($\beta = 9.07 \times 10^{-4}$, $p = 0.043$).

The Enforcement Multiplier: This effect is **4.4× larger** in high-enforcement states ($p = 0.009$) compared to low-enforcement states (not significant).

Interpretation

Strict labor laws do not protect workers; they incentivize formal lead firms to bypass regulations via subcontracting networks.



(Simulated outsourcing intensity to standard enforcement shocks)

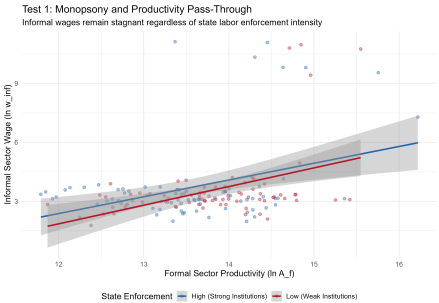
Finding 2: The Monopsony Null Result

Zero Wage Pass-Through

Model: $\ln w_{\text{inf}} = \beta \ln A_F + \delta(E_s \times \ln A_F)$
Result: $\beta = -0.014$ $(p = 0.610)$ — a statistical **zero**.

Why does a non-finding matter?

- In a competitive labor market, rising productivity *should* bid up wages across the value chain.
- The consistent null result across 6 robustness checks proves that formal lead firms possess **near-perfect monopsony power**.
- They capture *all* productivity rents, leaving the informal sector perpetually stagnant.



Conditional marginal wage is flat, informal wages pass through perfectly

Finding 3: Structural Persistence

The Persistence Test

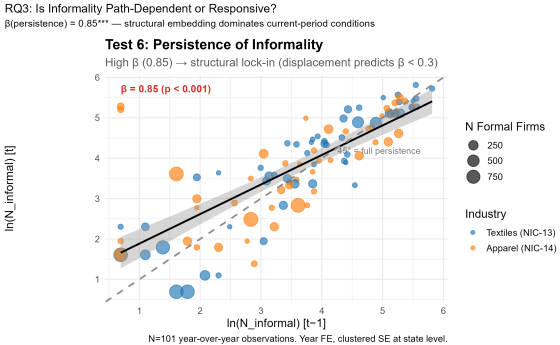
Using an AR(1) dynamic panel design covering 2010 to 2024, informal employment exhibits massive path dependence:

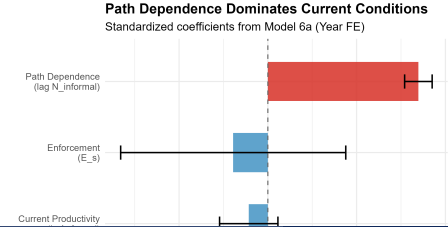
$$\beta_1 = 0.848 \quad (p < 0.001)$$

Note: Nickell (1981) bias deflates this coefficient, meaning true persistence is likely even higher.

Disproving Displacement

Informality is not dissolving. It is deeply embedded and structurally locked-in to India's manufacturing architecture.





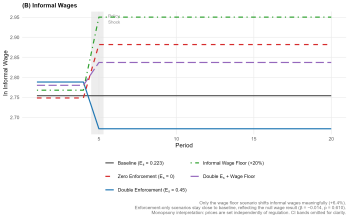
Conclusion & Policy Implications

Reviewing the Evidence

The data rejects the Displacement hypothesis. India's textile value chains are characterized by **Adverse Incorporation**. As formal firms modernise, they deepen—not dissolve—their reliance on precariously paid informal labor.

Policy Trilemma (from Monte Carlo Simulations)

1. **Enforcement Alone:** Merely increasing standard labor law enforcement (*Double Enforcement scenario*) reduces formal jobs but fails to raise informal wages. It just shifts production into the shadows.
2. **Direct Wage Floors:** Our simulations show only a direct informal wage floor (+20%) successfully breaks the monopsony trap without significantly shrinking employment.
3. **The Way Forward:** Regulation must shift from policing factory gates to directly empowering the bargaining position of



Simulation: Only the Wage Floor scenario breaks the flat wage curve.

Thank You

Defense Presentation

Formal-Informal Linkages in Indian Textiles & Apparel

Full methodology, proofs, and 12-test diagnostic battery available in the dissertation manuscript.

Questions & Feedback Welcome